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Unit 13 Hydrocarbons

BZX27Z

Learning Objectives

CHARLES ADOLPHE WURTZ

He is a French organic Chemist. He discovered phosphoryl chlorides. He showed that alkylhalides react with sodium to produce higher alkanes. This reaction was later named as Wurtz reaction. He is also known for his discoveries of ethylamine glycol and aldol condensation reactions.

After studying this unit, the students will be able to

- classify hydrocarbons according to nature of bond between the carbon atoms.
- name hydrocarbons according to IUPAC system of nomenclature.
- describe various methods of preparing hydrocarbon.
- interpret and use the following terminology in organic chemistry like free radicals, initiation, propagation and termination.
- describe chemistry of alkanes by the following reaction of halogenations, combustion, cracking.
- predict the formation of addition products of unsymmetrical alkenes which applies both Markovnikov's rule and peroxide effect.
- deduce acidic hydrogen in alkynes.
- comprehend the structure of benzene, explain aromaticity and understand mechanism of electrophilic substitutions reactions.
- predict the directive influence of substituents on mono substituted benzene.
- recognize the toxicity and carcinogenic nature of aromatic hydrocarbons.